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Section 14.2 : Gradient Vector, Tangent Planes And Normal Lines

In this section we want to revisit tangent planes only this time we'll look at them in light of the gradient vector. In the process we will also take a look at a normal line to a surface.

Let's first recall the equation of a plane that contains the point (x_0, y_0, z_0) with normal vector $\vec{n} = \langle a, b, c \rangle$ is given by,

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

When we introduced the gradient vector in the section on [directional derivatives](#) we gave the following fact.

Fact

The gradient vector $\nabla f(x_0, y_0)$ is orthogonal (or perpendicular) to the [level curve](#) $f(x, y) = k$ at the point (x_0, y_0) . Likewise, the gradient vector $\nabla f(x_0, y_0, z_0)$ is orthogonal to the level surface $f(x, y, z) = k$ at the point (x_0, y_0, z_0) .

Actually, all we need here is the last part of this fact. This says that the gradient vector is always orthogonal, or *normal*, to the surface at a point.

Also recall that the gradient vector is,

$$\nabla f = \langle f_x, f_y, f_z \rangle$$

So, the tangent plane to the surface given by $f(x, y, z) = k$ at (x_0, y_0, z_0) has the equation,

$$f_x(x_0, y_0, z_0)(x - x_0) + f_y(x_0, y_0, z_0)(y - y_0) + f_z(x_0, y_0, z_0)(z - z_0) = 0$$

This is a much more general form of the equation of a tangent plane than the one that we derived in the previous section.

Note however, that we can also get the equation from the previous section using this more general formula. To see this let's start with the equation $z = f(x, y)$ and we want to find the tangent plane to the surface given by $z = f(x, y)$ at the point (x_0, y_0, z_0) where $z_0 = f(x_0, y_0)$. In order to use the formula above we need to have all the variables on one side. This is easy enough to do. All we need to do is subtract a z from both sides to get,

$$f(x, y) - z = 0$$

Now, if we define a new function

$$F(x, y, z) = f(x, y) - z$$

we can see that the surface given by $z = f(x, y)$ is identical to the surface given by $F(x, y, z) = 0$ and this new equivalent equation is in the correct form for the equation of the tangent plane that we derived in this section.

So, the first thing that we need to do is find the gradient vector for F .

$$\nabla F = \langle F_x, F_y, F_z \rangle = \langle f_x, f_y, -1 \rangle$$

Notice that

$$\begin{aligned} F_x &= \frac{\partial}{\partial x}(f(x, y) - z) = f_x & F_y &= \frac{\partial}{\partial y}(f(x, y) - z) = f_y \\ F_z &= \frac{\partial}{\partial z}(f(x, y) - z) = -1 \end{aligned}$$

The equation of the tangent plane is then,

$$f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) - (z - z_0) = 0$$

Or, upon solving for z , we get,

$$z = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

which is identical to the equation that we derived in the previous section.

We can get another nice piece of information out of the gradient vector as well. We might on occasion want a line that is orthogonal to a surface at a point, sometimes called the **normal**

line. This is easy enough to get if we recall that the **equation of a line** only requires that we have a point and a parallel vector. Since we want a line that is at the point (x_0, y_0, z_0) we know that this point must also be on the line and we know that $\nabla f(x_0, y_0, z_0)$ is a vector that is normal to the surface and hence will be parallel to the line. Therefore, the equation of the normal line is,

$$\vec{r}(t) = \langle x_0, y_0, z_0 \rangle + t \nabla f(x_0, y_0, z_0)$$

Example 1 Find the tangent plane and normal line to $x^2 + y^2 + z^2 = 30$ at the point $(1, -2, 5)$.

Hide Solution ▼

For this case the function that we're going to be working with is,

$$F(x, y, z) = x^2 + y^2 + z^2$$

and note that we don't have to have a zero on one side of the equal sign. All that we need is a constant. To finish this problem out we simply need the gradient evaluated at the point.

$$\begin{aligned}\nabla F &= \langle 2x, 2y, 2z \rangle \\ \nabla F(1, -2, 5) &= \langle 2, -4, 10 \rangle\end{aligned}$$

The tangent plane is then,

$$2(x - 1) - 4(y + 2) + 10(z - 5) = 0$$

The normal line is,

$$\vec{r}(t) = \langle 1, -2, 5 \rangle + t \langle 2, -4, 10 \rangle = \langle 1 + 2t, -2 - 4t, 5 + 10t \rangle$$